

TACHS Exam Simulator

Full-Length Practice Exam 03

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: The Quiet Bell

Mara had volunteered to restore the brass bell from the town's abandoned ferry landing. She imagined polishing away a little tarnish and finishing before lunch. Instead, she found a cracked wooden handle, layers of green corrosion, and a clapper frozen in place.

At the workshop, Mr. Ibarra showed her how to loosen each part without damaging it. Mara wanted to hurry, but every shortcut left a scratch or bent an edge. She began keeping notes: which solution worked, how long each piece soaked, and where every washer belonged.

On the morning of the river festival, Mara fastened the repaired bell beside the new walking path. The mayor invited her to ring it. She pulled the handle, but only a dull knock sounded. Her face warmed as the crowd waited. Then Mara noticed that the clapper's leather strap had twisted during installation. She calmly removed the pin, straightened the strap, and tried again.

The clear note traveled over the water. Mara smiled, not because the bell sounded perfect, but because she now knew what to do when it did not.

1. Which statement best expresses the central idea of the passage?

- A) Careful work teaches Mara how to solve an unexpected problem.
- B) Public festivals should preserve objects from local history.
- C) Mr. Ibarra performs the most difficult part of the restoration.
- D) Old machinery usually works poorly after being repaired.

2. What caused the bell to make only a dull knock?

- A) The clapper was still corroded.
- B) A leather strap had become twisted.
- C) The wooden handle cracked again.
- D) A washer had been left at the workshop.

3. What can reasonably be inferred about Mara at the end of the passage?

- A) She regrets volunteering for the project.
- B) She believes the crowd did not notice the delay.
- C) She has become more patient and resourceful.
- D) She plans to replace the bell with a new one.

4. How does the final sentence contribute to the passage?

- A) It introduces a new conflict about the bell's condition.
- B) It explains why the festival was held beside the river.
- C) It reveals that Mara cares only about pleasing the crowd.
- D) It emphasizes that learning to respond to setbacks matters more than perfection.

Informational Passage: A Garden That Holds Rain

After a heavy storm, water rushing from roofs and pavement can overwhelm drains and carry soil or litter into streams. A rain garden offers a small-scale solution. It is a shallow planted area placed where runoff naturally collects. Rather than sending water immediately into a pipe, the garden briefly holds it.

A rain garden is not a permanent pond. Its soil is prepared to absorb water, and its surface usually dries within a day or two. Deep-rooted native plants are often suitable because their roots create channels in the soil. These plants may also require less watering after they become established.

Location matters. Builders avoid placing rain gardens directly against foundations or over underground utilities. They also test how quickly the soil drains. If water remains too long, mosquitoes may breed and plant roots may decay.

One garden cannot prevent every flood. However, many small gardens distributed through a neighborhood can reduce the amount of runoff reaching storm drains at the same moment. They can also provide food and shelter for insects and birds. Thus, a feature designed to manage water may improve local habitat as well.

5. According to the passage, why are deep-rooted plants useful in rain gardens?

- A) Their roots form channels that help water enter the soil.
- B) Their leaves prevent all rain from reaching the ground.
- C) Their flowers remove litter from nearby streams.
- D) Their stems strengthen building foundations.

6. As used in the final paragraph, “distributed” most nearly means

- A) sold for a profit
- B) spread among different places
- C) measured with precision
- D) removed from public land

7. The author’s primary purpose is to

- A) argue that storm drains should be eliminated
- B) describe the history of neighborhood landscaping
- C) explain how rain gardens work and why proper design matters
- D) compare the appearance of several native plants

8. Which conclusion is best supported by the passage?

- A) Every yard has soil suitable for a rain garden.
- B) Rain gardens remain filled with water throughout summer.
- C) A single rain garden can protect an entire city from flooding.
- D) The combined effect of many modest projects can be significant.

Humanities Passage: Reading a Building

People often treat buildings as backgrounds, noticing them only when searching for an entrance. Yet a building can be read as evidence of the choices and needs of the people who made it. A wide staircase may announce a public gathering place. A row of identical windows may reveal that builders valued efficiency, while carved stone above a doorway may show where they chose to spend extra labor.

Materials provide clues, too. Brick made from nearby clay can connect a structure to its region. A later wall built with different brick may mark an expansion. Even repairs tell a story: a patch can show both where damage occurred and what materials were available when it was fixed.

Interpreting these clues requires caution. A modern visitor might assume that a tiny window was intended for decoration, although it may have reduced heat loss. One feature can also serve several purposes. An awning can shelter a doorway, shade a room, and advertise a shop.

Looking closely does not mean every guess will be correct. It means replacing a quick impression with useful questions: What problem did this feature solve? Which parts are original? What changed over time? Buildings become more than scenery when observers recognize them as records shaped by climate, technology, money, and human intention.

9. Which sentence best summarizes the passage?

- A) Architectural features can provide evidence, but observers must interpret them thoughtfully.
- B) Old buildings should always be restored with local materials.
- C) Modern buildings contain fewer meaningful details than older ones.
- D) Decorative features are the most reliable clues to a building's purpose.

10. In the first paragraph, “announce” most nearly means

- A) publish
- B) shout
- C) signal
- D) celebrate

11. What might different brick in a later wall indicate?

- A) The building was expanded.
- B) The original clay was too colorful.
- C) The windows were once larger.
- D) The structure was moved to another region.

12. Why does the author mention that an awning can serve several purposes?

- A) To prove that shop signs are unnecessary
- B) To warn that a feature may not have one simple explanation
- C) To show that awnings are more useful than windows
- D) To recommend adding awnings to historic buildings

13. Which word best describes the author's tone?

- A) Alarmed
- B) Amused
- C) Doubtful
- D) Curious

Literary or Narrative Passage: The Washed Grid

Amaya and Tariq had spent two afternoons drawing a chalk grid across the brick wall behind the community center. The grid would guide a mural of birds crossing a sunrise, and each square corresponded to one square on their smaller sketch. By Friday evening, every line was in place.

A light rain arrived overnight. The wall beneath the roof edge stayed dry, but most of the grid faded into pale smudges. Tariq suggested beginning the mural anyway and judging the proportions by eye. Amaya looked at the wide blank wall, then at the detailed sketch. A small error in one corner, she realized, could grow into a crooked wing several feet away.

Three chalk marks had survived beneath the roof edge. Amaya measured the distance between them and compared it with the sketch. She and Tariq tied a string to the first mark, checked it with a level, and snapped fresh horizontal lines across the wall. Then they measured downward to rebuild the vertical lines.

The work delayed painting by an hour, but the new grid matched the original plan. At the unveiling, visitors noticed the bright birds, not the lines hidden beneath them. Amaya did. To her, the invisible grid was evidence that careful preparation could be rebuilt, even after it seemed to disappear.

14. Which parts of the original grid remained visible after the rain?

- A) Several vertical lines near the ground
- B) Three chalk marks beneath the roof edge
- C) The complete outline of one bird
- D) A row of squares beside the doorway

15. Why does Amaya decide not to judge the mural proportions by eye?

- A) She believes the wall is too wet to paint.
- B) She wants visitors to see the chalk lines.
- C) She has forgotten which colors belong in the design.
- D) She understands that a small measuring error could distort a large part of the mural.

16. Which lesson is best supported by Amaya's response to the erased grid?

- A) Careful methods can restore progress after an unexpected setback.
- B) Outdoor artwork should be completed only during dry seasons.
- C) A successful mural depends more on color than on measurement.
- D) Visitors appreciate artwork more when they understand every preparation step.

17. As used in the third paragraph, "tied a string to the first mark" most nearly shows that the string was

- A) measured loosely.
- B) hidden from view.
- C) secured at one point.
- D) colored with chalk.

Scientific or Technical Passage: How Gecko Feet Hold On

A gecko can race across a wall and pause on a ceiling without using glue, claws, or suction cups. The secret lies in structures too small to see without magnification. Each toe is covered with millions of microscopic hairs. Every hair branches into even finer tips that can make extremely close contact with a surface.

When those tips approach a wall, weak attractions form between molecules in the toe and molecules in the wall. One attraction is tiny, but millions acting together can support the gecko's weight. Because the hairs are flexible, they conform to small bumps and dips that would prevent a stiff surface from touching evenly.

Attachment alone would not be useful if the foot could not release. A gecko changes the angle of its toes as it steps. This motion peels the hair tips away in sequence, reducing their contact with the surface. The foot can therefore grip strongly in one position and release quickly in another.

Researchers study gecko feet while developing dry adhesives for robots and reusable fasteners. A successful material must do more than stick. It must also release on command, work repeatedly, and avoid leaving residue.

The gecko's foot demonstrates how many weak contacts, arranged and controlled carefully, can produce a strong but reversible grip.

18. What covers each gecko toe according to the passage?

- A) Millions of microscopic hairs
- B) A thin layer of natural glue
- C) Several flexible suction cups
- D) Rows of curved claws

19. Why does the author explain how a gecko changes the angle of its toes?

- A) To show why geckos move more slowly on ceilings
- B) To compare gecko toes with the claws of other animals
- C) To show that the grip can be released as well as created
- D) To argue that toe angle matters more than microscopic hairs

20. How is the passage mainly organized?

- A) It lists several animals and ranks their climbing ability.
- B) It presents competing theories and proves that one is false.
- C) It describes an invention and then traces its commercial history.
- D) It explains a natural mechanism, describes how it releases, and connects it to possible designs.

21. As used in the second paragraph, "conform" most nearly means

- A) become harder over time.
- B) fit closely to the shape of.
- C) move in a straight direction.
- D) produce a chemical coating.

Informational or Historical-Style Passage: When Screw Threads Became Predictable

A screw may look simple, but its spiral ridge has several important measurements. The diameter, the distance between ridges, and the shape of each ridge determine which nut will fit. In early workshops, craftspeople often cut threads with tools made for their own shop. A bolt produced in one place might not match a nut made elsewhere, even when both appeared to be the same size.

As machines became more widely used, this mismatch created practical problems. A damaged bolt could delay a repair while a replacement was filed or recut by hand. Merchants also had to store many nearly identical parts because customers used different thread patterns.

Manufacturers gradually adopted shared sets of thread sizes. Gauges allowed workers to check whether a new bolt matched an agreed diameter and spacing. Printed tables helped workshops select the same measurements, and suppliers could label parts more consistently.

The change required effort. Some shops replaced cutting tools, updated records, and trained workers to use the new gauges. Standardization did not make every screw identical; different jobs still required different sizes and strengths. It created recognized families of parts. Once those families were widely understood, repairs became more predictable, inventories became simpler, and parts made in distant workshops were more likely to fit together.

22. What did workers use to check whether a bolt matched an agreed thread size?

- A) Polishing stones
- B) Shipping labels
- C) Wooden patterns
- D) Gauges

23. What is the author's main purpose in the passage?

- A) To persuade readers that all machines should use one screw size
- B) To explain why shared thread standards developed and what problems they reduced
- C) To describe how one craftsman invented the first threaded bolt
- D) To compare the strength of metal and wooden fasteners

24. Which statement best summarizes the passage?

- A) Thread gauges eliminated the need for skilled workers in machine shops.
- B) Merchants preferred storing many similar parts because customers requested variety.
- C) Shared thread measurements made replacement parts and repairs more dependable across workshops.
- D) Standardization required every screw to have the same diameter and strength.

25. Which inference is best supported about workshops during the transition to shared standards?

- A) Some had to change tools and records before receiving the full benefits.
- B) Most stopped producing threaded parts until every shop agreed.
- C) They no longer needed to label replacement parts.
- D) They used fewer measurements because gauges replaced written tables.

Literary Passage: The Last Lantern

Mina had promised to bring a lantern to the neighborhood's evening garden walk. Her grandfather's brass lantern seemed perfect until she discovered that its glass door would not close. With an hour remaining, she carried it to Mr. Bell's repair shop.

Mr. Bell examined the bent latch. "I can straighten this," he said, "but brass remembers every impatient hand." Instead of reaching for a heavy tool, he warmed the latch beneath a small lamp and pressed it gradually with padded pliers. Mina watched the metal move by fractions.

"Wouldn't hitting it be faster?" she asked.

"For a moment," he replied. "Then we might spend tomorrow repairing what speed broke today."

When the latch finally clicked, Mina polished the cloudy glass. At the walk, she carried the lantern at the rear of the group. Its modest circle of light did not illuminate the entire path, but it revealed each stone before someone stepped on it. Mina stopped wishing she had brought the brightest lantern. She began matching her pace to the people ahead, keeping the light where it was needed.

The next morning, she returned the lantern to its shelf. Beside it, she placed a note: "Handle slowly. Useful things do not have to hurry."

26. Why does Mina visit Mr. Bell?

- A) The lantern's latch is bent.
- B) The lantern has run out of fuel.
- C) She wants to buy padded pliers.
- D) She needs directions to the garden.

27. What does Mr. Bell most likely mean when he says that "brass remembers every impatient hand"?

- A) Old objects can identify their owners.
- B) Brass becomes brighter when handled.
- C) Careless force can leave lasting damage.
- D) Repairs should always be completed by hand.

28. As used in the passage, "modest" most nearly means

- A) newly repaired.
- B) limited but adequate.
- C) difficult to notice.
- D) unusually beautiful.

29. Which statement best expresses a central idea of the passage?

- A) Antique objects should be displayed rather than used.
- B) The brightest tool is generally the most dependable.
- C) Community events require careful advance planning.
- D) Patience can protect useful things and help people serve others well.

Literary Passage: A Different Route

The trail map showed one straight path to the overlook, and Eli had memorized every turn. Halfway up the hill, however, a fallen oak blocked the narrow trail. Its branches stretched into thorny brush on both sides. Eli's cousin Nia pointed to a faint path descending toward the creek. "The maintenance crew may use that route," she said. Eli disliked walking downhill when their destination was above them, but climbing over the unstable trunk was clearly unsafe.

They followed the faint path. Near the creek, the trail disappeared beneath smooth stones. Nia noticed three small blue marks painted on rocks at regular intervals. The marks led upstream, crossed a wooden footbridge, and joined the main trail beyond the fallen tree.

At the overlook, Eli unfolded his map. The maintenance route was absent. He sketched it lightly in the margin, including the bridge and blue markers, but added a question mark beside the entrance. On the return trip, they met a ranger, who confirmed that the route was open temporarily while crews cleared the oak.

Eli erased the question mark and wrote "temporary detour." He had always treated maps as finished answers. Now he understood them as records that sometimes needed careful updates from the ground beneath his feet.

30. What helps Eli and Nia follow the route near the creek?

- A) Blue marks painted on rocks
- B) Arrows carved into trees
- C) Directions printed on the map
- D) Signs posted by the ranger

31. Why does Eli first place a question mark beside the route he sketches?

- A) He cannot remember where the route begins.
- B) He is uncertain whether the route is approved for hikers.
- C) He plans to ask Nia to redraw the bridge.
- D) He believes the blue marks indicate danger.

32. How does the author mainly develop Eli's changing view of maps?

- A) By comparing maps produced in several periods
- B) By describing the ranger's daily responsibilities
- C) By presenting a problem that experience helps Eli reconsider
- D) By explaining every step used to print a trail map

33. Which word best describes the author's tone?

- A) Playful
- B) Alarmed
- C) Disapproving
- D) Thoughtful

Informational Passage: Why Some Cities Build Rain Gardens

During a storm, rain falling on roofs, sidewalks, and parking lots cannot soak directly into soil. Instead, it flows across these hard surfaces, gathering leaves, dirt, and other materials before entering street drains. A large volume arriving quickly can burden drainage systems and carry pollutants into streams.

A rain garden is a shallow planted basin designed to receive some of this runoff. Water enters the basin, spreads among grasses and flowering plants, and slowly filters through layers of soil. The garden is not a permanent pond. Designers select its depth, soil mixture, and drainage area so that standing water usually disappears within a day or two.

Plant choice matters. Deep-rooted native species can loosen soil and tolerate alternating wet and dry periods.

Their stems also slow moving water. Yet a rain garden must be placed carefully. It should not sit over underground utilities or directly against a building foundation, and its size must match the area draining into it.

Rain gardens cannot prevent every flood or remove every pollutant. Their value comes from handling many smaller flows before those flows combine. When numerous properties use them, the gardens function like a scattered network of small sponges, reducing pressure on a city's drainage system while creating habitat for insects and birds.

34. According to the passage, why are deep-rooted native plants useful in rain gardens?

- A) They loosen soil and tolerate changing moisture levels.
- B) They prevent all water from reaching street drains.
- C) They require the basin to remain permanently flooded.
- D) They protect underground utilities from damage.

35. As used in the first paragraph, “burden” most nearly means

- A) repair.
- B) conceal.
- C) place excessive demand on.
- D) carry beneath the ground.

36. Which statement best summarizes the passage?

- A) Rain gardens replace city drains by holding storm water permanently.
- B) Rain gardens use carefully chosen sites, soils, and plants to slow and filter runoff, though they have limits.
- C) Native plants survive only when they grow near building foundations.
- D) Most storm pollution comes from soil inside planted basins.

37. Why does the author compare many rain gardens to “a scattered network of small sponges”?

- A) To show that each garden must be made from an artificial material
- B) To suggest that rain gardens make city soil permanently wet
- C) To explain why insects build homes beneath the plants
- D) To emphasize their combined ability to absorb and slow water in many locations

38. What is the author's main purpose?

- A) To explain how rain gardens work and identify both their benefits and limitations
- B) To persuade readers that every lawn should be replaced by a pond
- C) To criticize cities that maintain underground drainage systems
- D) To provide instructions for identifying every native plant species

Literary Passage: The Unclaimed Apron

When the community kitchen closed after Saturday's bread workshop, Nia noticed a blue apron hanging behind the pantry door. Its pocket held a card covered with careful notes about oven temperatures. She assumed the apron belonged to Mr. Han, the instructor, but he had already left.

Nia could have placed it in the lost-and-found bin. Instead, she studied the card. One note mentioned that the old oven's back corner heated fastest. Another warned that its timer rang two minutes late. The information would matter during Sunday's charity bake sale, when volunteers unfamiliar with the kitchen would prepare dozens of loaves.

Nia called Mr. Han. The apron was not his; it belonged to Mrs. Ortiz, who had tested the oven all week. Mrs. Ortiz asked Nia to leave both items with the morning coordinator. Before going home, Nia copied the two most important warnings onto a large sheet and taped it beside the oven controls.

The next morning, volunteers rotated their pans and used a separate timer. By noon, every loaf had baked evenly. Mrs. Ortiz thanked Nia for returning the apron, then smiled at the sign. "You returned more than what you found," she said.

39. What was inside the apron pocket?

- A) A list of volunteers
- B) A card with oven notes
- C) A separate timer
- D) A bread recipe

40. Why did Nia make a large sign?

- A) She wanted to advertise another workshop.
- B) She needed to identify the apron's owner.
- C) She hoped to replace the oven's controls.
- D) She wanted all volunteers to see the useful warnings.

41. In the final sentence, Mrs. Ortiz means that Nia "returned more than what you found" because Nia

- A) shared the apron's useful information with others.
- B) brought extra supplies into the kitchen.
- C) returned several objects from the lost-and-found bin.
- D) taught the volunteers how to bake bread.

42. Which statement best expresses the passage's central idea?

- A) Old equipment should always be replaced before a large event.
- B) Written directions are more dependable than experienced instructors.
- C) Acting thoughtfully on a small discovery can help an entire group.
- D) Community events succeed only when one person takes charge.

Scientific Passage: Measuring a Tree Without Climbing It

Foresters often need to estimate a tree's height without climbing it. One simple method uses shadows and similar triangles. On a sunny day, a person places a straight measuring stick upright on level ground near the tree. The stick and tree cast shadows at the same time. Because sunlight reaches both objects at the same angle, the two objects and their shadows form triangles with matching angles.

The forester measures the stick, its shadow, and the tree's shadow. Suppose a one-meter stick casts a two-meter shadow while a tree casts a thirty-meter shadow. The tree is fifteen times taller than the stick because its shadow is fifteen times longer. Its estimated height is therefore fifteen meters.

Accuracy depends on careful conditions. The stick must stand vertically, and both shadows should fall across ground with a similar slope. Blurred shadow tips can make endpoints difficult to judge. Wind may also move branches, changing the apparent end of the tree's shadow. Taking several measurements and averaging the results can reduce the effect of small errors.

This method does not provide a perfect measurement, but it allows foresters to gather useful information quickly with ordinary tools.

43. Why do the tree and stick form triangles with matching angles?

- A) They stand on exactly the same spot.
- B) Their shadows always have equal lengths.
- C) Sunlight reaches both at the same angle.
- D) Foresters measure them with the same tool.

44. How is the passage mainly organized?

- A) It presents a claim and then argues against it.
- B) It explains a method, gives an example, and discusses sources of error.
- C) It compares two professions and recommends one.
- D) It lists events in the history of forest measurement.

45. Why does the author mention taking several measurements?

- A) To explain how averaging can reduce small measurement errors
- B) To prove that one measurement can never be useful
- C) To suggest that different trees cast identical shadows
- D) To show that the stick should be moved after every reading

46. Which statement best summarizes the passage?

- A) Tree height can be found exactly whenever the sun is visible.
- B) Foresters prefer shadows because measuring tools are unreliable.
- C) Wind is the main reason trees are difficult to measure.
- D) Shadow proportions offer a practical estimate when measurements are taken carefully.

Biographical-Style Passage: Designing for Many Hands

As a young craft worker, Elena Varga repaired household tools in a busy neighborhood shop. She noticed that many customers struggled with ordinary objects for reasons manufacturers had overlooked. A jar opener might require a strong grip. A measuring cup might have markings visible from only one side. Instead of blaming users, Varga began asking how the objects could change.

Years later, as a product designer, she invited people of different ages and abilities to test early models. Their comments often altered her plans. When testers had difficulty turning a narrow handle, she made it wider and added a textured surface. When left-handed users covered measurement labels while pouring, she printed labels on both sides.

Some colleagues worried that these changes would make products complicated. Varga argued that thoughtful features could simplify an object for nearly everyone. A broad handle helped a person with limited grip strength, but it also helped a cook whose hands were wet. Labels on both sides assisted left-handed users and anyone viewing the cup from an unusual angle.

Varga's greatest contribution was not a single invention. It was her habit of treating difficulty as information. By observing where people hesitated, she discovered opportunities to design objects that welcomed more users.

47. What change did Varga make after testers struggled with a narrow handle?

- A) She removed the handle.
- B) She added measurement labels.
- C) She made it wider and textured.
- D) She replaced the early model with a jar.

48. Which belief most clearly guided Varga's work?

- A) Products should be tested only by trained designers.
- B) A user's difficulty can reveal a weakness in a design.
- C) Specialized products should serve only one group.
- D) Attractive objects are usually the easiest to use.

49. In the final paragraph, "welcomed" most nearly means

- A) greeted aloud.
- B) entertained briefly.
- C) sold directly to.
- D) made usable and inviting for.

50. What is the author's primary purpose?

- A) To show how attentive observation helped a designer create more usable objects
- B) To explain why handmade tools are superior to manufactured ones
- C) To argue that all products should have identical handles
- D) To describe the steps for becoming a professional craft worker

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: Borrowing More Than Books

[1] Our school library recently began lending small tool kits. [2] Each kit contains basic items such as a screwdriver, measuring tape, and level. [3] Students may borrow one for three days with a library card. [4] The idea began after a survey showed that many families occasionally needed tools but did not want to purchase them. [5] The kits were popular right away. [6] During the first month, every kit was borrowed at least twice. [7] The library also displays novels near the front desk. [8] Because the program is useful, the student council plans to collect donations for additional kits.

51. Which sentence should be removed because it does not support the passage's focus?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 7
- D) Sentence 8

52. Which revision most effectively combines sentences 5 and 6?

- A) The kits were popular right away, during the first month.
- B) The kits were immediately popular; every kit was borrowed at least twice during the first month.
- C) Popular right away, and every kit was borrowed at least twice during the first month.
- D) Every kit was borrowed, the kits were popular at least twice during the first month.

53. Which word best replaces "useful" in sentence 8 to make the statement more precise?

- A) successful
- B) nice
- C) thing-like
- D) okay

54. Which sentence would provide the most effective conclusion?

- A) Libraries have existed for a very long time.
- B) Some screwdrivers have flat tips, while others have crossed tips.
- C) Surveys can contain questions with several answer choices.
- D) With more kits available, even more families can complete occasional repairs without buying rarely used tools.

Revision Passage: Preparing for the Night Walk

[1] Next Friday, the science club will lead an evening nature walk in Pine Hollow Park. [2] Participants should bring a flashlight and wear sturdy shoes. [3] The group will pause at three marked locations to listen for frogs and insects. [4] Club members has practiced identifying each species by its call. [5] Before the walk begins, a park educator will explain how artificial light can affect nocturnal animals. [6] Participants will then cover their flashlights with red filters, which are less disruptive than bright white beams. [7] Everyone must remain on the trail, speak quietly, and to follow the leader. [8] These steps will help the group observe wildlife responsibly.

55. Which revision corrects sentence 4?

- A) Club members have practiced identifying each species by its call.
- B) Club members having practiced identifying each species by its call.
- C) Club members has been practicing to identify each species by its call.
- D) Club members is practiced identifying each species by its call.

56. Which revision makes the series in sentence 7 parallel?

- A) Everyone must remain on the trail, speaking quietly, and follows the leader.
- B) Everyone must remain on the trail, quiet speech, and following the leader.
- C) Everyone must remain on the trail, speak quietly, and follow the leader.
- D) Everyone must remaining on the trail, speak quietly, and to follow the leader.

57. Which revision of sentence 3 is most concise without changing its meaning?

- A) The group, which is the group participating, will make pauses at three locations that are marked.
- B) At three marked locations, there will be pauses made by the group to listen.
- C) The group will pause and stop at three marked locations for listening purposes.
- D) The group will stop at three marked locations to listen for frogs and insects.

58. Where should this sentence be placed?

“This brief lesson will prepare walkers to limit their impact on the park.”

- A) After sentence 2
- B) After sentence 5
- C) After sentence 7
- D) After sentence 8

59. Which sentence is written correctly?

- A) Neither of the two routes lead directly to the museum.
- B) The box of donated maps is beside the entrance.
- C) Each of the volunteers were given a name tag.
- D) One of those lanterns need a new battery.

60. Which choice is a complete sentence?

- A) Although the rehearsal ended later than expected.
- B) Carrying the painted scenery into the auditorium.
- C) Because the musicians needed one final practice.
- D) The stage manager checked every microphone before the doors opened.

61. Which revision is clearest?

“When Lena spoke with Ava, she said the schedule had changed.”

- A) When Lena spoke with Ava, she mentioned it.
- B) Lena told Ava that the schedule had changed.
- C) Speaking with Ava, the schedule had changed according to Lena.
- D) She told her about the changed schedule when they spoke.

62. Which line contains an error in standard written US English?

Line 1: The robotics team tested its new design on Tuesday.

Line 2: Every wheel turned smoothly during the first trial.

Line 3: Later, the students adjusted the sensors careful.

Line 4: They recorded the results in a shared notebook.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

63. Which line contains an error in standard written US English?

Line 1: Maya and me presented the proposal to the committee.

Line 2: We explained how the new signs would help visitors.

Line 3: Several members asked thoughtful questions.

Line 4: The committee approved the plan unanimously.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: A Textile Collection Day

[1] Our environmental club organized a textile collection day for clothing and household fabric that families no longer used. [2] The goal was to keep usable cloth in circulation and damaged fabric out of the trash. [3] We placed labeled bins for wearable items, torn cotton, and mixed fabric beside the gym entrance. [4] Volunteers checked each bag before placing its contents in the correct bin. [5] Volunteers weighed each group before packing it. [6] A local shelter accepted the wearable items, while a recycling company took fabric that could not be reused. [7] By the end of the day, the bins held 186 pounds of material. [8] The project showed that clear sorting instructions can turn unwanted fabric into useful resources.

64. Which sentence would best follow sentence 2 to develop the explanation of the collection plan?

- A) The club had held several meetings in the same gym during the year.
- B) Some volunteers wore blue shirts, while others wore green shirts.
- C) To support that goal, the club separated wearable items from fabric that required recycling.
- D) The collection day began on a Saturday morning in early spring.

65. Which revision of sentence 5 is most precise?

- A) Volunteers weighed the fabric in each category before packing it into labeled boxes.
- B) Volunteers did something with every group before putting it away.
- C) Before packing, the groups were weighed by volunteers in a way.
- D) Volunteers weighed them before they packed all of it.

66. Which revision most effectively combines sentences 1 and 2?

- A) Our environmental club organized a textile collection day, families no longer used clothing and household fabric.
- B) The goal was to keep usable cloth in circulation, our environmental club organized a collection day.
- C) Our environmental club organized a textile collection day and the goal, keeping fabric out of the trash.
- D) To keep usable cloth in circulation and damaged fabric out of the trash, our environmental club organized a textile collection day.

67. Which sentence would best replace sentence 8 as a conclusion?

- A) Some families brought fabric in paper bags, while others used boxes.
- B) Because the bins were clearly labeled, families sorted materials quickly and helped redirect 186 pounds of fabric.
- C) The club may choose a different gym entrance for its next event.
- D) Wearable clothing is available in many colors and sizes.

Revision Passage: Testing Parcel Padding

[1] Our engineering class compared three recycled materials for protecting a fragile object inside a package. [2] Each team placed the same type of clay cup in an identical box. [3] Using shredded paper, folded cardboard, or crumpled cloth, the boxes were packed. [4] We dropped every box from a height of one meter onto the same rubber mat. [5] Before each trial, the cups were checked for cracks. [6] The cloth-packed boxes protected all five cups, while the other materials protected four cups each. [7] The classroom windows faced east. [8] We concluded that cloth performed best in our small test, but more trials would be needed before making a broad claim.

68. Where should the following sentence be placed? Each padding material was tested in five separate boxes.

- A) Before sentence 1
- B) After sentence 3
- C) After sentence 6
- D) After sentence 8

69. Which revision best corrects the misplaced modifier in sentence 3?

- A) Using shredded paper, folded cardboard, or crumpled cloth, identical boxes were available.
- B) The boxes, using shredded paper, folded cardboard, or crumpled cloth, were packed by materials.
- C) The teams packed the boxes using shredded paper, folded cardboard, or crumpled cloth.
- D) Shredded paper, folded cardboard, or crumpled cloth, the teams packed the boxes.

70. Which revision of sentence 6 states the result most precisely?

- A) All five cups packed with cloth remained uncracked, compared with four of five cups in each other group.
- B) The cloth worked better, while the other materials were almost as good somehow.
- C) Most cups survived, and cloth was a material used in some boxes.
- D) The boxes with cloth had cups, but the other boxes also had cups that were protected.

71. Which revision of sentence 5 uses consistent past tense and a clear subject?

- A) Before each trial, the cups are checked for cracks.
- B) Before each trial, checking the cups for cracks.
- C) Before each trial, the cups had checked for cracks.
- D) Before each trial, we checked each cup for cracks.

72. Which sentence is written correctly?

- A) A box of old cables were stored beneath the table.
- B) A box of old cables have been stored beneath the table.
- C) A box of old cables was stored beneath the table.
- D) A box of old cables store beneath the table.

73. Which sentence uses pronouns correctly?

- A) The teacher gave Luis and I the updated schedule.
- B) The teacher gave Luis and me the updated schedule.
- C) The teacher gave he and Luis the updated schedule.
- D) The teacher gave Luis and myself the updated schedule.

74. Which line contains an error in standard written US English?

Line 1: The volunteers carried the labeled bins outside.

Line 2: Each bin was placed beside the correct truck.

Line 3: The coordinator checked the final count.

Line 4: Neither of the labels were attached securely.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

75. Which line contains an error in standard written US English?

Line 1: The donations arrived before noon.

Line 2: Two students recorded the weight of each bag.

Line 3: The volunteers sorted the materials careful.

Line 4: Later, the team cleaned the collection area.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

76. Which line contains an error in standard written US English?

Line 1: Each of the posters have a date printed at the bottom.

Line 2: The largest poster hangs beside the entrance.

Line 3: Visitors can read the schedule from the hallway.

Line 4: A smaller copy is posted near the stairs.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Recording Local Sounds

[1] Our media club decided to create an audio map of the neighborhood. [2] Teams recorded short sounds at the market, train platform, park, and riverside path. [3] We used microphones to make recordings of sounds that could be heard. [4] Back at school, students labeled each clip with its location and time. [5] Because wind covered several voices, those interviews had to be recorded again. [6] Next, the club arranged the clips to guide listeners from the busiest place to the quietest. [7] The cafeteria serves vegetable soup on Wednesdays. [8] A brief spoken introduction explains why familiar places can sound different at different hours. [9] The completed map will be posted on the school website.

77. Which revision of sentence 3 is most concise?

- A) We made recordings with microphones of sounds that people were able to hear.
- B) Microphones were the equipment used by us for audible recordings.
- C) We recorded the sounds with microphones.
- D) Using microphones, sounds that could be heard were recorded by our club.

78. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 7
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

79. Which revision most effectively combines sentences 4 and 5?

- A) Back at school, students labeled each clip, and its location and time, wind covered voices.
- B) After students labeled each clip with its location and time, they rerecorded interviews in which wind had covered voices.
- C) Students labeled clips with locations and times, because recording the interviews again.
- D) Labeling every clip back at school, several voices covered by wind and repeated.

80. Which phrase best replaces “the busiest place to the quietest” in sentence 6 to make the comparison precise?

- A) one place and then another place
- B) locations with a whole lot of noise toward calmer ones
- C) the places that were busy through those not busy
- D) the location with the loudest activity to the location with the least

Revision Passage: The Seed Library

[1] A seed library allows gardeners to borrow packets of seeds and later donate seeds they collect. [2] Our community center opened one in a wooden cabinet near the entrance. [3] Volunteers sorted packets by plant type, they also added labels showing planting depth and season. [4] Visitors can choose seeds without paying, but they are encouraged to return some after a successful harvest. [5] This exchange preserves locally successful varieties. [6] It also gives beginning gardeners an inexpensive way to experiment. [7] Clear instructions beside the cabinet explain how to dry and label donated seeds. [8] The seed library is small, yet each returned envelope helps the collection continue.

81. Which revision correctly repairs sentence 3?

- A) Volunteers sorted packets by plant type and adding labels that showed planting depth and season.
- B) Volunteers sorted packets by plant type; they also added labels showing planting depth and season.
- C) Volunteers sorted packets by plant type, also labels showing planting depth and season.
- D) Volunteers sorting packets by plant type while labels showed planting depth and season.

82. Where should the following sentence be placed?

Some packets contain beans and herbs that families have grown in the area for decades.

- A) After sentence 5
- B) After sentence 2
- C) Before sentence 1
- D) After sentence 8

83. Which revision of sentence 8 is most effective?

- A) Though modest in size, the seed library grows stronger with every properly returned envelope.
- B) The seed library is little, but envelopes are things that make it go.
- C) Small is the seed library, and returning is continuing it.
- D) The cabinet has envelopes, yet it is a small kind of library.

84. Which sentence is written correctly?

- A) Each of the painted signs have a protective coating.
- B) Neither the coach nor the players was expecting rain.
- C) The collection of old postcards belong in this drawer.
- D) Every one of the volunteers receives a name tag.

85. Which sentence uses the pronoun correctly?

- A) The librarian gave Luis and I the cabinet key.
- B) Between you and me, the final clue was difficult.
- C) Her and Malik finished the display before noon.
- D) The judges congratulated she after the contest.

86. Which sentence has correct parallel structure?

- A) The internship required arriving early, careful notes, and to ask questions.
- B) Our goals were to reduce waste, saving money, and cleaner rooms.
- C) The guide taught us how to measure, record, and compare the samples.
- D) Nina enjoys sketching buildings, to photograph doors, and maps.

87. Which line contains an error in standard written US English?

Line 1: The museum guide spoke clearly.

Line 2: She described how the first machine worked.

Line 3: The visitors listened attentive to every detail.

Line 4: Afterward, they asked several questions.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

88. Which line contains an error in standard written US English?

Line 1: Before the doors opened, Marcus checked the tickets.

Line 2: He placed each stack on the correct table.

Line 3: The signs were already hanging above the entrances.

Line 4: By nine o'clock, every volunteers had arrived.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: A Hallway Exhibition

[1] Our art club planned an exhibition for the long hallway outside the auditorium. [2] Because students hurry through that space between classes. [3] We decided that each display needed to communicate its idea quickly. [4] Members selected twelve pieces and wrote brief labels explaining the materials and inspiration behind them. [5] The labels were printed in large type so that visitors could read them from a comfortable distance. [6] The auditorium curtains are dark red. [7] On installation day, we arranged the works at consistent heights and checked that no frame blocked a doorway. [8] By afternoon, the hallway had become an orderly gallery that students could enjoy without slowing traffic.

89. Which revision best combines sentences 2 and 3 into one complete sentence?

- A) Because students hurry through that space between classes, we decided that each display needed to communicate its idea quickly.
- B) Students hurry through that space between classes, because deciding each display needed to communicate quickly.
- C) Because students hurry through that space between classes; each display needed to communicate its idea quickly.
- D) Students hurrying through that space between classes and each display communicating quickly.

90. Which sentence should be removed because it does not support the passage's focus?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 6
- D) Sentence 7

91. Which revision of sentence 4 is most concise while preserving its meaning?

- A) Members made a selection of twelve different pieces and then proceeded to write labels that were brief.
- B) Twelve pieces were selected by members, who wrote labels about various things concerning them.
- C) Members selected pieces, and the pieces had materials as well as inspiration that labels explained.
- D) Members chose twelve works and wrote brief labels about their materials and inspiration.

92. Which sentence would best follow sentence 5?

- A) Several club members also enjoy painting landscapes at home.
- B) We also spaced the works apart so viewers could distinguish one display from the next.
- C) The school auditorium hosts concerts during the winter.
- D) Some printed materials are produced by large machines.

Revision Passage: Testing the Solar Cooker

[1] Our science team built a solar cooker from a box, reflective panels, and a clear cover. [2] We wanted to learn how the angle of the panels affected the temperature inside the box. [3] At ten-minute intervals, Jamal recorded the temperature, checking the position of the sun. [4] The first trial reached only 38°C. [5] For the second trial, we adjusted the panels so more sunlight was directed through the cover. [6] The temperature then rose to 51°C under the same outdoor conditions. [7] The second arrangement was more better at concentrating sunlight. [8] Our results showed that careful panel placement can substantially improve the cooker's performance.

93. Which revision best corrects the misplaced modifier in sentence 3?

- A) Checking the sun's position, the temperature was recorded by Jamal at ten-minute intervals.
- B) Jamal, at ten-minute intervals checking, recorded the temperature and the sun's position.
- C) At ten-minute intervals, Jamal checked the sun's position and recorded the temperature.
- D) The sun's position checked, at ten-minute intervals Jamal recorded the temperature.

94. Which revision correctly replaces “more better” in sentence 7?

- A) best
- B) better
- C) most better
- D) more good

95. Which revision of sentence 6 most clearly emphasizes the comparison between trials?

- A) The temperature was 51°C, and outdoor conditions existed.
- B) Under the same outdoor conditions, the second trial reached 51°C, thirteen degrees higher than the first.
- C) It then rose under conditions that were the same outside to a temperature of 51°C.
- D) The second temperature, which was a temperature, rose to 51°C.

96. Which sentence uses verb tense correctly?

- A) By the time the bus arrived, we had waited for twenty minutes.
- B) By the time the bus arrived, we wait for twenty minutes.
- C) By the time the bus arrives yesterday, we had waited for twenty minutes.
- D) By the time the bus arrived, we are waiting for twenty minutes.

97. Which sentence uses pronouns correctly?

- A) The coach spoke privately with him and me.
- B) Her and Elena designed the poster.
- C) The award was presented to Marcus and I.
- D) Us students carried the boxes inside.

98. Which line contains an error in standard written US English?

Line 1: The collection of old photographs was stored safely.

Line 2: Each image had been placed in a protective sleeve.

Line 3: A volunteer carefully recorded its date and source.

Line 4: Later, the curator organize the images by neighborhood.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

99. Which line contains an error in standard written US English?

Line 1: The two routes lead to the same overlook.

Line 2: This trail is shorter than the path beside the lake.

Line 3: However, it climbs most steeply than the other route.

Line 4: Both paths are clearly marked.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

100. Which line contains an error in standard written US English?

Line 1: Before sunrise, the crew loaded the equipment.

Line 2: They secured every container with a strap.

Line 3: The driver inspected the tires and mirrors.

Line 4: Then the trucks moves slowly toward the gate.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. What is $\frac{3}{5}$ of 45?

- A) 15
- B) 27
- C) 30
- D) 75

102. A jacket priced at \$64 is discounted by 25%. What is the sale price?

- A) \$16
- B) \$48
- C) \$39
- D) \$56

103. Which value is greatest?

- A) 0.62
- B) 61%
- C) $\frac{5}{8}$
- D) 0.6

104. Solve $4x - 7 = 21$.

- A) 7
- B) 14
- C) 16
- D) 28

105. The sequence begins 6, 11, 16, 21. Which expression gives the n th term?

- A) $5n - 1$
- B) $5n + 1$
- C) $6n - 1$
- D) $6n$

106. If $y = 3x + 2$, what is y when $x = -4$?

- A) -14
- B) -12
- C) -10
- D) 10

107. The numbers of pages read on five days were 18, 22, 15, 25, and 20. What was the mean?

- A) 20
- B) 18
- C) 21
- D) 25

108. A bag contains 4 blue, 3 green, and 5 yellow tiles. One tile is selected, replaced, and then another is selected. What is the probability that both tiles are green?

- A) $\frac{1}{16}$
- B) $\frac{1}{12}$
- C) $\frac{1}{9}$
- D) $\frac{1}{4}$

109. A triangle has angles measuring 48° and 67° . What is the measure of the third angle?

- A) 55°
- B) 60°
- C) 65°
- D) 75°

110. A rectangle has area 84 square centimeters and length 12 centimeters. What is its perimeter?

- A) 19 cm
- B) 42 cm
- C) 84 cm
- D) 38 cm

111. A recipe uses 750 milliliters of broth. How many liters is this?

- A) 0.075 L
- B) 7.5 L
- C) 75 L
- D) 0.75 L

112. A bus leaves at 9:35 a.m. and arrives at 12:10 p.m. How long is the trip?

- A) 2 hours 25 minutes
- B) 3 hours 25 minutes
- C) 3 hours 35 minutes
- D) 2 hours 35 minutes

113. What is the value of $2\frac{1}{3} \div \frac{7}{9}$?

- A) $1\frac{5}{9}$
- B) $2\frac{1}{9}$
- C) 3
- D) $4\frac{2}{3}$

114. A supply room has 84 pencils and 126 erasers. What is the greatest number of identical kits that can be made using all the items?

- A) 21
- B) 42
- C) 84
- D) 126

115. Which expression has a value of -14 ?

- A) $-3 \times 6 + 4$
- B) $18 \div (-3) + 4$
- C) $-20 - (-8)$
- D) $5 - 3 \times 4$

116. Which inequality is the solution to $3x + 5 \leq 20$?

- A) $x \geq 5$
- B) $x < 5$
- C) $x > 5$
- D) $x \leq 5$

117. A function has the values $x = 1, 3, 5$ and $y = 4, 10, 16$. Which rule describes the function?

- A) $y = 2x + 2$
- B) $y = 3x + 1$
- C) $y = 3x - 1$
- D) $y = 4x$

118. The six measurements are 14, 18, 19, 21, 26, and 32. What is the median?

- A) 19
- B) 19.5
- C) 20
- D) 21

119. A spinner has eight equal sectors labeled A, A, A, B, B, C, D, and D. What is the probability of landing on A?

- A) $\frac{3}{8}$
- B) $\frac{1}{2}$
- C) $\frac{5}{8}$
- D) $\frac{3}{5}$

120. Of 120 students surveyed, 42 ride a bus, 30 walk, and 18 ride a bicycle. The rest travel by car. What percent travel by car?

- A) 20%
- B) 22.5%
- C) 24%
- D) 25%

121. What is the distance between the points $(-3, 4)$ and $(5, 4)$?

- A) 2 units
- B) 6 units
- C) 8 units
- D) 12 units

122. Two triangles are similar. A side of 6 centimeters on the smaller triangle corresponds to a side of 10 centimeters on the larger triangle. If another side of the smaller triangle is 9 centimeters, what is the corresponding side of the larger triangle?

- A) 12 centimeters
- B) 13.5 centimeters
- C) 14 centimeters
- D) 15 centimeters

123. A trapezoid has parallel bases of 8 centimeters and 14 centimeters and a height of 5 centimeters. What is its area?

- A) 44 square centimeters
- B) 55 square centimeters
- C) 70 square centimeters
- D) 110 square centimeters

124. A roll contains 4 yards 2 feet of ribbon. How many feet of ribbon are on the roll?

- A) 14 feet
- B) 12 feet
- C) 10 feet
- D) 6 feet

125. A package has a mass of 2.4 kilograms. An additional item has a mass of 650 grams. What is the combined mass in kilograms?

- A) 2.465 kilograms
- B) 2.95 kilograms
- C) 3.00 kilograms
- D) 3.05 kilograms

126. A jacket priced at \$64 is discounted 25%. What is the sale price?

- A) \$16
- B) \$39
- C) \$48
- D) \$52

127. What is the value of $\frac{3}{5} + \frac{7}{10} \div 2$?

- A) $\frac{19}{20}$
- B) $\frac{13}{20}$
- C) $\frac{4}{5}$
- D) $1\frac{3}{10}$

128. A recipe uses flour and oats in the ratio 5:2. If 21 cups of the mixture are prepared, how many cups are oats?

- A) 3
- B) 5
- C) 15
- D) 6

129. The first four terms of a sequence are 4, 9, 19, 39. Which rule produces each term after the first?

- A) Add 5.
- B) Multiply by 2, then add 1.
- C) Multiply by 3, then subtract 3.
- D) Add consecutive multiples of 5.

130. Solve $4(2x - 3) = 3x + 18$.

- A) $x = 3$
- B) $x = 5$
- C) $x = 6$
- D) $x = 8$

131. A streaming service charges \$7 per month plus \$2 for each movie rented. Which expression gives the cost for m movies in one month?

- A) $7m + 2$
- B) $9m$
- C) $7(m + 2)$
- D) $7 + 2m$

132. The numbers of pages read by five students are 18, 22, 22, 25, and 33. What is the mean?

- A) 24
- B) 22
- C) 25
- D) 120

133. A bag contains 3 red, 2 blue, and 5 green tiles. One tile is drawn, replaced, and a second tile is drawn. What is the probability that both tiles are blue?

- A) $\frac{1}{10}$
- B) $\frac{1}{20}$
- C) $\frac{1}{25}$
- D) $\frac{2}{25}$

134. Two angles of a triangle measure 47° and 68° . What is the measure of the third angle?

- A) 55°
- B) 65°
- C) 75°
- D) 115°

135. A rectangle has an area of 96 square centimeters and a length of 12 centimeters. What is its perimeter?

- A) 40 cm
- B) 32 cm
- C) 20 cm
- D) 48 cm

136. A tank contains 3.5 liters of water. How many milliliters is this?

- A) 35
- B) 350
- C) 30,500
- D) 3,500

137. A bus leaves at 10:47 a.m. and arrives at 1:18 p.m. How long is the trip?

- A) 2 hours 41 minutes
- B) 2 hours 31 minutes
- C) 3 hours 29 minutes
- D) 3 hours 31 minutes

138. A tank is $\frac{3}{5}$ full. After 18 liters are added, it is $\frac{9}{10}$ full. What is the tank's total capacity?

- A) 45 liters
- B) 54 liters
- C) 72 liters
- D) 60 liters

139. What is the value of $4.8 \div 0.6 + 3 \times 2$?

- A) 11
- B) 14
- C) 16
- D) 22

140. A club's membership increased from 80 students to 92 students. What was the percent increase?

- A) 15%
- B) 12%
- C) 18%
- D) 20%

141. The perimeter of a rectangle is 46 centimeters. Its length is 5 centimeters more than its width. What is the width?

- A) 8 cm
- B) 9 cm
- C) 13 cm
- D) 18 cm

142. A machine produces 7 finished parts every 12 minutes. At the same constant rate, how many parts does it produce in 60 minutes?

- A) 28
- B) 30
- C) 35
- D) 42

143. The numbers of tickets sold on four days were 36, 42, 39, and 47. How many tickets must be sold on the fifth day for the five-day mean to be 42?

- A) 42
- B) 44
- C) 48
- D) 46

144. A bag contains 4 red, 5 blue, and 3 green tiles. One tile is selected, not replaced, and then a second tile is selected. What is the probability that both tiles are green?

- A) $\frac{1}{11}$
- B) $\frac{1}{12}$
- C) $\frac{1}{22}$
- D) $\frac{1}{44}$

145. Data Set P is 6, 8, 10, 12, 14. Data Set Q is 2, 8, 10, 12, 18. Which statement is true?

- A) Both sets have the same mean, but Q has a greater range.
- B) P has a greater mean and a greater range.
- C) Q has a greater mean, but both sets have the same range.
- D) Both sets have the same mean and the same range.

146. Each exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

- A) 8
- B) 7
- C) 6
- D) 9

147. A circle has a radius of 6 centimeters. Using $\pi = 3.14$, what is its area?

- A) 37.68 cm^2
- B) 75.36 cm^2
- C) 113.04 cm^2
- D) 226.08 cm^2

148. A rectangular prism measures 8 cm by 5 cm by 3 cm. What is its total surface area?

- A) 79 cm^2
- B) 120 cm^2
- C) 240 cm^2
- D) 158 cm^2

149. A cyclist travels 18 miles in 1.5 hours. At the same rate, how far will the cyclist travel in 2 hours?

- A) 20 miles
- B) 22 miles
- C) 24 miles
- D) 27 miles

150. A recipe requires 250 milliliters of broth per batch. How many liters are needed for 6 batches?

- A) 0.15 L
- B) 1.5 L
- C) 15 L
- D) 150 L

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Complete the 2×2 matrix.

Top-left: ○

Top-right: ○○

Bottom-left: ●

Bottom-right: ?

Across each row, one matching shape is added. Down each column, open shapes become filled.

- A) ●●
- B) ○●
- C) ■■
- D) ●●●

152. Complete the 2×2 matrix.

Top-left: ↑

Top-right: →

Bottom-left: ←

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise.

- A) →
- B) ↓
- C) ↑
- D) ←

153. Complete the 3×3 matrix.

Row 1: ○ | □ | ○□

Row 2: △ | ◇ | △◇

Row 3: ☆ | + | ?

In each row, the third cell combines the first two cells in the same order.

- A) +☆
- B) ☆+
- C) ★+
- D) ☆×

154. Complete the 2×2 matrix.

Top-left: open shape ○ above filled shape ■

Top-right: filled shape ■ above open shape ○

Bottom-left: open shape △ above filled shape ◆

Bottom-right: ?

Across each row, the top and bottom positions switch.

- A) open △ above filled ◆
- B) filled ▲ above open ◇
- C) open ◇ above filled ▲
- D) filled ◆ above open △

155. Complete the 3×3 matrix.

Row 1: ○ | ● | ○

Row 2: ● | ○ | ●

Row 3: ○ | ● | ?

The cells alternate between open and filled circles across every row and column.

- A) ●
- B) ○
- C) □
- D) ■

156. A square paper is folded in half from left to right. One hole is punched near the upper-right corner of the folded rectangle. After unfolding, where are the holes?

- A) Near both lower corners
- B) At the centers of the left and right edges
- C) Near both upper corners
- D) One near the upper-right corner only

157. A square paper is folded bottom to top, then left to right. One hole is punched at the exact center of the final folded square. After both folds are undone, what results?

- A) Two holes along the vertical centerline
- B) Four holes, one in each quadrant, symmetric across both centerlines
- C) Four holes along the top edge
- D) Eight holes around the border

158. A rectangular paper is folded in half from top to bottom. A triangular notch is cut from the folded edge, away from both corners. What appears after unfolding?

- A) A diamond-shaped opening centered on the horizontal centerline
- B) Two separate notches, one on each outer edge
- C) One triangular notch on the top edge
- D) A circular opening at the center

159. A square paper is folded along the diagonal from upper-left to lower-right. One hole is punched in the folded triangle away from the diagonal and all edges. After unfolding, what results?

- A) Two holes reflected across that diagonal
- B) Two holes reflected across the vertical centerline
- C) Four holes reflected across both centerlines
- D) One hole on the diagonal

160. The figures ○●○, □■□, and △▲△ share a rule. Which choice follows the same rule?

- A) ◆◆◆
- B) ★☆☆
- C) ◇◆◆
- D) ●○○

161. The arrangements “○ inside □,” “△ inside ◇,” and “☆ inside ○” share a rule. Which choice follows the same rule?

- A) ■ inside ◆
- B) + inside ×
- C) ● beside □
- D) ◇ inside △

162. The figures $\uparrow \rightarrow \downarrow$, $\rightarrow \downarrow \leftarrow$, and $\downarrow \leftarrow \uparrow$ share a rule. Which choice follows the same rule?

- A) $\uparrow \downarrow \rightarrow$
- B) $\leftarrow \downarrow \rightarrow$
- C) $\rightarrow \leftarrow \downarrow$
- D) $\leftarrow \uparrow \rightarrow$

163. Complete the 2×3 matrix.

Row 1: ○○○○ | ○ | ●●●

Row 2: △△△△△ | △△ | ?

In each row, subtract the second count from the first count, then change the remaining open shapes to filled shapes.

- A) △△△
- B) ▲▲
- C) △△
- D) ▲▲▲

164. Complete the 2×2 matrix.

Top-left: Top ○; Bottom □

Top-right: Left ○; Right □

Bottom-left: Top △; Bottom ◇

Bottom-right: ?

Across each row, the vertical arrangement turns 90° counterclockwise without changing symbol order.

- A) Left ◇; Right △
- B) Left △; Right ◇
- C) Top ◇; Bottom △
- D) Top △; Bottom ◇

165. Complete the 3×3 matrix.

Row 1: ○ | ●● | ○○○

Row 2: □□ | ■■■ | □

Row 3: △△△ | ▲ | ?

The counts 1, 2, and 3 shift one position left in each new row. The center cell in every row is filled.

- A) △△
- B) ▲▲
- C) △
- D) ▲▲▲

166. Complete the 2×2 matrix.

Top-left: ○□

Top-right: ●□

Bottom-left: ○■

Bottom-right: ?

Moving right fills only the first shape. Moving down fills only the second shape.

A) ○□

B) ●□

C) ●■

D) ○■

167. A rectangular sheet is divided into three equal vertical panels. The left panel is folded right onto the center panel, and the right panel is folded left onto the same center panel. One hole is punched near the top of the final stack, away from every edge and fold. What appears after unfolding?

A) Two holes in a horizontal row

B) Three holes in a horizontal row, one in each panel

C) Three holes in a vertical column

D) Six holes arranged in two rows

168. A square sheet is folded once from left to right. Two holes are punched in the folded rectangle: one near the upper-right corner and one near the lower-right corner, both away from every edge and the fold. What appears after unfolding?

A) Two holes near the right edge

B) Two holes along the vertical centerline

C) Three holes forming a triangle

D) Four holes, one near each corner

169. A rectangular sheet is folded from top to bottom. The folded rectangle is then divided into three equal vertical panels; the left panel is folded onto the center, and the right panel is folded onto the center. One interior hole is punched through the final stack. What appears after unfolding?

A) Three holes in one horizontal row

B) Four holes in a 2×2 arrangement

C) Six holes in a 2×3 arrangement

D) Eight holes in two rows

170. A rectangular sheet is folded in half from left to right. A semicircular notch is cut at the midpoint of the folded packet's right outer edge, away from the fold. What appears after unfolding?

A) Two matching semicircular notches, one at the midpoint of each outer vertical edge

B) One circular hole at the center of the sheet

C) Two notches along the top edge

D) Four semicircular notches, one near each corner

171. The following arrangements share the same rule:

○ □ ■ ●

△ ◇ ◆ ▲

☆ ○ ● ★

Which choice follows the same rule?

A) □ △ ▲ ■

B) □ △ ■ ▲

C) ■ ▲ △ □

D) □ ▲ △ ■

172. The following arrangements share the same rule:

○ ■■

△ ◆◆

☆ ●●

Which choice follows the same rule?

A) □ □□

B) ■ △△

C) □ ▲ ▲

D) ◆ ○○

173. The following arrangements share the same rule:

↑ ○ ●

→ □ ■

↓ △ ▲

Which choice follows the same rule?

A) ← ◆ ◇

B) ← ◇ □

C) ← ■ □

D) ← ◇ ◆

174. The following groups share the same rule:

| - /

/ | -

- / |

Which choice follows the same rule?

A) | | /

B) | / -

C) - - |

D) / \ /

175. Complete the matrix.

Row 1: ○ | ●

Row 2: □ | ?

Across each row, the shape keeps its form and changes from open to filled.

A) ○

B) ■

C) ●

D) □

176. Complete the matrix.

Row 1: ↑ | → | ↓

Row 2: → | ↓ | ←

Row 3: ↓ | ← | ?

Each step to the right or downward rotates the arrow 90° clockwise.

A) ↑

B) →

C) ←

D) ↓

177. Complete the matrix.

Row 1: ○ | ○○

Row 2: △△ | ?

Across each row, one matching shape is added.

A) △△△

B) ○○○

C) △

D) ▲▲▲

178. Complete the matrix.

Row 1: ○□ | □○

Row 2: △◇ | ?

Across each row, the left and right positions exchange.

A) △◇

B) ◇△

C) ▲◆

D) ◇◇

179. Complete the matrix.

Row 1: ○ | □ | □○

Row 2: △ | ☆ | ?

In each row, the third cell combines the first two symbols in order.

A) ☆△

B) △△

C) △☆

D) ○□

180. A square sheet is folded from top to bottom. One hole is punched near the upper-right corner of the folded rectangle, away from the fold. What appears after unfolding?

A) Four holes, one near each corner

B) Two holes near the upper-left and upper-right corners

C) One hole on the horizontal centerline

D) Two holes near the upper-right and lower-right corners

181. A square sheet is folded from right to left, then from top to bottom. A hole is punched near the lower-left corner of the final folded square, away from both folds. What appears after complete unfolding?

A) Two holes reflected across only the vertical centerline

B) Four holes, one in each quadrant, reflected across both centerlines

C) Two holes reflected across only the horizontal centerline

D) Eight holes arranged around the center

182. A rectangular sheet is folded once along its vertical centerline. A triangular cut is made on the folded edge, above the midpoint. What appears after unfolding?

A) One diamond-shaped opening above the center

B) Two triangular notches on the outer edges

C) Four triangular openings around the center

D) One triangular opening on the left side

183. A square is folded along the diagonal from upper-left to lower-right. A hole is punched near the upper-right corner of the folded triangle, away from the fold. What appears after unfolding?

- A) One hole at the center
- B) Four holes symmetric across both diagonals
- C) Two holes mirrored across the upper-left-to-lower-right diagonal
- D) Two holes on the vertical centerline

184. The three arrangements share the same rule.

1: ○ ● ○

2: □ ■ □

3: △ ▲ △

Which choice follows the same rule?

- A) ○ □ ○
- B) ◇ ◆ ◇
- C) ◆ ◇ ◆
- D) ☆ ☆ ★

185. The three arrangements share the same rule.

1: ↑ → ↓

2: → ↓ ←

3: ↓ ← ↑

Which choice follows the same rule?

- A) ← ↓ ↑
- B) ↑ ↓ →
- C) → ↑ ←
- D) ← ↑ →

186. The three figures share the same rule.

1: ○ inside □; □ inside △

2: △ inside ◇; ◇ inside ☆

3: ☆ inside ○; ○ inside □

Which choice follows the same rule?

- A) ○ inside ○; ○ inside △
- B) ◆ inside □; □ outside △
- C) □ inside △; △ inside ◇
- D) △ outside ◇; ◇ inside ☆

187. The three arrangements share the same rule.

1: Top-left ○; top-right □; bottom-left □; bottom-right ○

2: Top-left △; top-right ◇; bottom-left ◇; bottom-right △

3: Top-left ☆; top-right +; bottom-left +; bottom-right ☆

Which choice follows the same rule?

- A) Top-left ○; top-right □; bottom-left ○; bottom-right □
- B) Top-left △; top-right △; bottom-left ◇; bottom-right ◇
- C) Top-left +; top-right ×; bottom-left +; bottom-right ×
- D) Top-left ●; top-right ■; bottom-left ■; bottom-right ●

188. Complete the 2×2 matrix.

Top-left: ○ above □

Top-right: □ above ○

Bottom-left: ● above ■

Bottom-right: ?

- A) ● above ■
- B) ■ above ●
- C) □ above ○
- D) ■ above ○

189. Complete the 2×2 matrix.

Top-left: ○

Top-right: ○+

Bottom-left: □□

Bottom-right: ?

- A) □+
- B) □□
- C) □□+
- D) +□□+

190. Complete the 3×3 matrix.

Row 1: ○ | □ | ○□

Row 2: △ | ☆ | △☆

Row 3: ◇ | + | ?

- A) ◇+
- B) +◇
- C) ◇◇
- D) ++

191. Complete the 2×2 matrix.

Top-left: Top ○; Bottom ●

Top-right: Top ●; Bottom ○

Bottom-left: Left □; Right ■

Bottom-right: ?

- A) Left □; Right ■
- B) Top ■; Bottom □
- C) Left ○; Right ●
- D) Left ■; Right □

192. A square sheet is folded in half from left to right. The folded rectangle is then folded from bottom to top. One hole is punched near the folded shape's upper-right corner. What appears when the sheet is fully unfolded?

- A) Two holes near the right corners
- B) Four holes, one near each corner
- C) Four holes along the vertical centerline
- D) Two holes along the top edge

193. A square sheet is folded once along the diagonal from upper-left to lower-right. A small triangular cut is made on the folded diagonal, below the center. What appears after unfolding?

- A) One diamond-shaped opening below the center
- B) Two separate triangular holes near the left edge
- C) Four triangular holes around the center
- D) One triangular opening above the center

194. A rectangular sheet is folded from right to left along its vertical centerline. One circular hole is punched exactly on the folded sheet's horizontal centerline, near its new right edge. What appears after unfolding?

- A) One hole at the center
- B) Two holes mirrored across the horizontal centerline
- C) Two holes mirrored across the vertical centerline
- D) Four holes mirrored across both centerlines

195. A square sheet is folded from top to bottom, then from right to left. A semicircular cut is made on the folded shape's bottom edge, away from all corners and fold intersections. What appears after unfolding?

- A) Two circular openings
- B) Four semicircular notches on the outer edges
- C) Two diamond-shaped openings
- D) Two matching circular openings reflected across the vertical centerline

196. Three arrangements follow the same rule:

○ | ●

□ | ■

△ | ▲

Which choice follows the same rule?

- A) ◇ | ◆
- B) ◆ | ◇
- C) ☆ | ○
- D) ★ | ★

197. Three arrangements follow the same rule:

↑ → ↓

→ ↓ ←

↓ ← ↑

Which choice follows the same rule?

- A) ← ↓ →
- B) ↑ ← ↓
- C) ← ↑ →
- D) → ↑ ←

198. Three arrangements follow the same rule:

○ inside □; ● outside □

△ inside ◇; ▲ outside ◇

☆ inside ○; ★ outside ○

Which choice follows the same rule?

- A) ◆ inside △; ◇ outside △
- B) □ inside ☆; ■ outside ☆
- C) ■ inside ○; □ outside ○
- D) ○ inside ○; ● outside □

199. Three arrangements follow the same rule:

Top: ○; **Center:** □; **Bottom:** ○

Top: △; **Center:** ☆; **Bottom:** △

Top: ◇; **Center:** +; **Bottom:** ◇

Which choice follows the same rule?

A) **Top:** ○; **Center:** ●; **Bottom:** □

B) **Top:** +; **Center:** ○; **Bottom:** ×

C) **Top:** ☆; **Center:** ☆; **Bottom:** +

D) **Top:** ■; **Center:** □; **Bottom:** ■

200. Three groups follow the same rule:

○ □ △ ○

□ △ ○ □

△ ○ □ △

Which choice follows the same rule?

A) ○ □ ○ △

B) □ □ △ □

C) △ □ ○ △

D) ☆ + × ☆

Answer Key

Question numbers continue across all five sections.

Reading

1. A	2. B	3. C	4. D	5. A	6. B	7. C	8. D
9. A	10. C	11. A	12. B	13. D	14. B	15. D	16. A
17. C	18. A	19. C	20. D	21. B	22. D	23. B	24. C
25. A	26. A	27. C	28. B	29. D	30. A	31. B	32. C
33. D	34. A	35. C	36. B	37. D	38. A	39. B	40. D
41. A	42. C	43. C	44. B	45. A	46. D	47. C	48. B
49. D	50. A						

Written Expression

51. C	52. B	53. A	54. D	55. A	56. C	57. D	58. B
59. B	60. D	61. B	62. C	63. A	64. C	65. A	66. D
67. B	68. B	69. C	70. A	71. D	72. C	73. B	74. D
75. C	76. A	77. C	78. A	79. B	80. D	81. B	82. A
83. A	84. D	85. B	86. C	87. C	88. D	89. A	90. C
91. D	92. B	93. C	94. B	95. B	96. A	97. A	98. D
99. C	100. D						

Mathematics

101. B	102. B	103. C	104. A	105. B	106. C	107. A	108. A
109. C	110. D	111. D	112. D	113. C	114. B	115. A	116. D
117. B	118. C	119. A	120. D	121. C	122. D	123. B	124. A
125. D	126. C	127. A	128. D	129. B	130. C	131. D	132. A
133. C	134. B	135. A	136. D	137. B	138. D	139. B	140. A
141. B	142. C	143. D	144. C	145. A	146. A	147. C	148. D
149. C	150. B						

Ability

151. A	152. C	153. B	154. D	155. B	156. C	157. B	158. A
159. A	160. C	161. D	162. D	163. D	164. B	165. A	166. C
167. B	168. D	169. C	170. A	171. A	172. C	173. D	174. B
175. B	176. A	177. A	178. B	179. C	180. D	181. B	182. A
183. C	184. B	185. D	186. C	187. D	188. B	189. C	190. A
191. D	192. B	193. A	194. C	195. D	196. A	197. C	198. B

199. D	200. D						
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Detailed Explanations

Reading

1. TACHS-R3-001 - Answer A: Careful work teaches Mara how to solve an unexpected problem.
Mara's patient restoration work gives her the knowledge and confidence to correct the final problem.
2. TACHS-R3-002 - Answer B: A leather strap had become twisted.
Mara notices that the clapper's leather strap twisted during installation.
3. TACHS-R3-003 - Answer C: She has become more patient and resourceful.
Mara calmly diagnoses and repairs the problem instead of rushing or giving up.
4. TACHS-R3-004 - Answer D: It emphasizes that learning to respond to setbacks matters more than perfection.
The final sentence contrasts a perfect result with Mara's newly developed problem-solving ability.
5. TACHS-R3-005 - Answer A: Their roots form channels that help water enter the soil.
The passage directly states that deep roots create channels in the soil.
6. TACHS-R3-006 - Answer B: spread among different places
The gardens are located throughout the neighborhood rather than concentrated in one place.
7. TACHS-R3-007 - Answer C: explain how rain gardens work and why proper design matters
The passage describes their function, benefits, placement, and drainage requirements.
8. TACHS-R3-008 - Answer D: The combined effect of many modest projects can be significant.
Many neighborhood rain gardens together can reduce simultaneous runoff into storm drains.
9. TACHS-R3-009 - Answer A: Architectural features can provide evidence, but observers must interpret them thoughtfully.
The passage presents buildings as records while warning readers against quick assumptions.
10. TACHS-R3-010 - Answer C: signal
A wide staircase can indicate or signal that the building is a public gathering place.
11. TACHS-R3-011 - Answer A: The building was expanded.
The passage says different brick in a later wall may mark an expansion.
12. TACHS-R3-012 - Answer B: To warn that a feature may not have one simple explanation
The awning example demonstrates that one architectural feature can serve multiple functions.
13. TACHS-R3-013 - Answer D: Curious
The author encourages close observation and thoughtful questions about architectural clues.
14. TACHS-R3-014 - Answer B: Three chalk marks beneath the roof edge
The passage states that three chalk marks survived in the dry area beneath the roof edge.
15. TACHS-R3-015 - Answer D: She understands that a small measuring error could distort a large part of the mural.
Amaya realizes that a minor error on the sketch scale could become a large distortion on the wall.
16. TACHS-R3-016 - Answer A: Careful methods can restore progress after an unexpected setback.
Amaya and Tariq recover from the erased grid by using surviving evidence and careful measurement.
17. TACHS-R3-017 - Answer C: secured at one point.
The string is fixed to the first mark so it can guide a straight replacement line.
18. TACHS-R3-018 - Answer A: Millions of microscopic hairs
The first paragraph identifies millions of microscopic hairs as the structures covering each toe.
19. TACHS-R3-019 - Answer C: To show that the grip can be released as well as created
Changing toe angle peels the hair tips away, showing how the gecko controls release after gripping.
20. TACHS-R3-020 - Answer D: It explains a natural mechanism, describes how it releases, and connects it to possible designs.
The passage moves from attachment to controlled release and then to engineering applications.
21. TACHS-R3-021 - Answer B: fit closely to the shape of.
Flexible hairs conform by fitting closely around tiny bumps and dips in a surface.
22. TACHS-R3-022 - Answer D: Gauges
The passage states that gauges checked whether bolts matched agreed diameters and spacing.
23. TACHS-R3-023 - Answer B: To explain why shared thread standards developed and what problems they reduced
The passage explains the mismatch problem, the adoption process, and the practical benefits of standards.

24. TACHS-R3-024 - Answer C: Shared thread measurements made replacement parts and repairs more dependable across workshops.
Recognized thread families made parts more interchangeable without requiring every screw to be identical.
25. TACHS-R3-025 - Answer A: Some had to change tools and records before receiving the full benefits.
The passage notes that shops replaced tools, updated records, and trained workers during adoption.
26. TACHS-R3-026 - Answer A: The lantern's latch is bent.
Mina seeks repair because the bent latch prevents the glass door from closing.
27. TACHS-R3-027 - Answer C: Careless force can leave lasting damage.
Mr. Bell warns that striking the brass could create additional damage.
28. TACHS-R3-028 - Answer B: limited but adequate.
The lantern lights only a small area, yet that light is sufficient to reveal stones.
29. TACHS-R3-029 - Answer D: Patience can protect useful things and help people serve others well.
Careful repair preserves the lantern, and Mina later uses its light thoughtfully.
30. TACHS-R3-030 - Answer A: Blue marks painted on rocks
Regularly spaced blue marks guide the cousins upstream toward the footbridge.
31. TACHS-R3-031 - Answer B: He is uncertain whether the route is approved for hikers.
Eli waits for the ranger's confirmation before labeling the route an official temporary detour.
32. TACHS-R3-032 - Answer C: By presenting a problem that experience helps Eli reconsider
The blocked trail forces Eli to discover that maps may require careful updates.
33. TACHS-R3-033 - Answer D: Thoughtful
The passage calmly reflects on how direct experience changes Eli's understanding.
34. TACHS-R3-034 - Answer A: They loosen soil and tolerate changing moisture levels.
The third paragraph directly identifies both benefits of deep-rooted native plants.
35. TACHS-R3-035 - Answer C: place excessive demand on.
A sudden large volume of runoff can overwhelm or strain drainage systems.
36. TACHS-R3-036 - Answer B: Rain gardens use carefully chosen sites, soils, and plants to slow and filter runoff, though they have limits.
This choice includes the passage's method, design requirements, benefits, and limitations.
37. TACHS-R3-037 - Answer D: To emphasize their combined ability to absorb and slow water in many locations
The sponge comparison illustrates the distributed gardens' collective effect on runoff.
38. TACHS-R3-038 - Answer A: To explain how rain gardens work and identify both their benefits and limitations
The author objectively describes their operation, placement, value, and limits.
39. TACHS-R3-039 - Answer B: A card with oven notes
The passage states that the apron pocket contained a card of notes about oven temperatures.
40. TACHS-R3-040 - Answer D: She wanted all volunteers to see the useful warnings.
Nia enlarged the two warnings so Sunday's unfamiliar volunteers could use them.
41. TACHS-R3-041 - Answer A: shared the apron's useful information with others.
Nia returned the apron and also made its valuable information available to the entire group.
42. TACHS-R3-042 - Answer C: Acting thoughtfully on a small discovery can help an entire group.
Nia's careful response to one found object prevents baking problems for many volunteers.
43. TACHS-R3-043 - Answer C: Sunlight reaches both at the same angle.
Equal sunlight angles produce similar triangles for the stick and tree.
44. TACHS-R3-044 - Answer B: It explains a method, gives an example, and discusses sources of error.
The passage progresses from procedure to numerical example and then to accuracy concerns.
45. TACHS-R3-045 - Answer A: To explain how averaging can reduce small measurement errors
The author directly identifies repeated measurements and averaging as ways to limit minor errors.
46. TACHS-R3-046 - Answer D: Shadow proportions offer a practical estimate when measurements are taken carefully.
This choice captures both the method's usefulness and its need for careful conditions.
47. TACHS-R3-047 - Answer C: She made it wider and textured.
Varga widened the narrow handle and added texture after observing testers.
48. TACHS-R3-048 - Answer B: A user's difficulty can reveal a weakness in a design.
Varga treats hesitation as evidence that an object may need improvement.

49. TACHS-R3-049 - Answer D: made usable and inviting for.

The objects welcomed more users by becoming easier and more comfortable for them to use.

50. TACHS-R3-050 - Answer A: To show how attentive observation helped a designer create more usable objects

The biography emphasizes Varga's observation-based approach and its benefits for varied users.

Written Expression

51. TACHS-WE3-001 - Answer C: Sentence 7

The location of novels does not develop the passage's focus on the tool-lending program.

52. TACHS-WE3-002 - Answer B: The kits were immediately popular; every kit was borrowed at least twice during the first month.

The semicolon correctly joins two related independent clauses without creating a run-on.

53. TACHS-WE3-003 - Answer A: successful

"Successful" precisely connects the program's demonstrated demand to the plan for expansion.

54. TACHS-WE3-004 - Answer D: With more kits available, even more families can complete occasional repairs without buying rarely used tools.

This conclusion reinforces both the planned expansion and the program's central benefit.

55. TACHS-WE3-005 - Answer A: Club members have practiced identifying each species by its call.

The plural subject "members" requires the plural helping verb "have."

56. TACHS-WE3-006 - Answer C: Everyone must remain on the trail, speak quietly, and follow the leader.

All three items use parallel base-form verbs governed by "must."

57. TACHS-WE3-007 - Answer D: The group will stop at three marked locations to listen for frogs and insects.

This version preserves every essential detail without redundancy or awkward passive phrasing.

58. TACHS-WE3-008 - Answer B: After sentence 5

"This brief lesson" clearly refers to the educator's explanation in sentence 5.

59. TACHS-WE3-009 - Answer B: The box of donated maps is beside the entrance.

The singular subject "box" correctly agrees with the singular verb "is."

60. TACHS-WE3-010 - Answer D: The stage manager checked every microphone before the doors opened.

This choice contains a subject, a complete predicate, and no dependent-word fragment.

61. TACHS-WE3-011 - Answer B: Lena told Ava that the schedule had changed.

Naming Lena as the speaker removes the unclear pronoun reference.

62. TACHS-WE3-012 - Answer C: Line 3

The verb "adjusted" requires the adverb "carefully," not the adjective "careful."

63. TACHS-WE3-013 - Answer A: Line 1

The compound subject requires the subject pronoun "I," as in "Maya and I."

64. TACHS-WE3-014 - Answer C: To support that goal, the club separated wearable items from fabric that required recycling.

The added sentence directly connects the stated goal with the sorting system described next.

65. TACHS-WE3-015 - Answer A: Volunteers weighed the fabric in each category before packing it into labeled boxes.

The revision identifies what was weighed and how the sorted fabric was packed.

66. TACHS-WE3-016 - Answer D: To keep usable cloth in circulation and damaged fabric out of the trash, our environmental club organized a textile collection day.

The introductory purpose phrase clearly and correctly connects the goal with the club's action.

67. TACHS-WE3-017 - Answer B: Because the bins were clearly labeled, families sorted materials quickly and helped redirect 186 pounds of fabric.

The revision connects the sorting method with the project's measured result and central purpose.

68. TACHS-WE3-018 - Answer B: After sentence 3

The sentence belongs after the materials are introduced and before the drop procedure and results.

69. TACHS-WE3-019 - Answer C: The teams packed the boxes using shredded paper, folded cardboard, or crumpled cloth.

The revision makes "the teams" the clear performers of the action described by the opening phrase.

70. TACHS-WE3-020 - Answer A: All five cups packed with cloth remained uncracked, compared with four of five cups in each other group.

The revision gives exact results for the cloth group and both comparison groups.

71. TACHS-WE3-021 - Answer D: Before each trial, we checked each cup for cracks.

"We checked" provides a clear subject and matches the passage's past-tense narration.

72. TACHS-WE3-022 - Answer C: A box of old cables was stored beneath the table.

The singular subject "box" requires the singular verb "was."

73. TACHS-WE3-023 - Answer B: The teacher gave Luis and me the updated schedule.
The object of “gave” requires the objective pronoun “me.”
74. TACHS-WE3-024 - Answer D: Line 4
The singular subject “neither” requires “was,” not “were.”
75. TACHS-WE3-025 - Answer C: Line 3
The adverb “carefully,” not the adjective “careful,” must modify “sorted.”
76. TACHS-WE3-026 - Answer A: Line 1
The singular subject “each” requires “has,” not “have.”
77. TACHS-WE3-027 - Answer C: We recorded the sounds with microphones.
This revision expresses the original idea directly without redundant references to audible sounds.
78. TACHS-WE3-028 - Answer A: Sentence 7
The cafeteria menu is unrelated to creating the neighborhood audio map.
79. TACHS-WE3-029 - Answer B: After students labeled each clip with its location and time, they rerecorded interviews in which wind had covered voices.
The revision clearly subordinates the first action and connects both completed tasks.
80. TACHS-WE3-030 - Answer D: the location with the loudest activity to the location with the least
The revision states the intended comparison precisely and consistently.
81. TACHS-WE3-031 - Answer B: Volunteers sorted packets by plant type; they also added labels showing planting depth and season.
A semicolon correctly joins the two closely related independent clauses.
82. TACHS-WE3-032 - Answer A: After sentence 5
The example of long-grown plants directly supports the claim about preserving locally successful varieties.
83. TACHS-WE3-033 - Answer A: Though modest in size, the seed library grows stronger with every properly returned envelope.
This revision is concise, clear, and appropriately emphasizes the effect of returned seeds.
84. TACHS-WE3-034 - Answer D: Every one of the volunteers receives a name tag.
The singular subject “every one” correctly takes the singular verb “receives.”
85. TACHS-WE3-035 - Answer B: Between you and me, the final clue was difficult.
The preposition “between” requires the objective pronouns “you” and “me.”
86. TACHS-WE3-036 - Answer C: The guide taught us how to measure, record, and compare the samples.
All three coordinated actions use parallel base-form verbs.
87. TACHS-WE3-037 - Answer C: Line 3
The adverb “attentively,” not the adjective “attentive,” must modify “listened.”
88. TACHS-WE3-038 - Answer D: Line 4
The singular determiner “every” requires “volunteer,” not the plural “volunteers.”
89. TACHS-WE3-039 - Answer A: Because students hurry through that space between classes, we decided that each display needed to communicate its idea quickly.
The dependent clause is correctly joined to an independent clause with a comma.
90. TACHS-WE3-040 - Answer C: Sentence 6
The curtain color does not support the passage’s focus on planning and installing the hallway exhibition.
91. TACHS-WE3-041 - Answer D: Members chose twelve works and wrote brief labels about their materials and inspiration.
This revision preserves all essential information while eliminating unnecessary wording.
92. TACHS-WE3-042 - Answer B: We also spaced the works apart so viewers could distinguish one display from the next.
The sentence adds a relevant design choice that supports clear viewing.
93. TACHS-WE3-043 - Answer C: At ten-minute intervals, Jamal checked the sun’s position and recorded the temperature.
Jamal clearly performs both actions, eliminating the misplaced modifier.
94. TACHS-WE3-044 - Answer B: better
The standard comparative form of “good” is “better,” without “more.”
95. TACHS-WE3-045 - Answer B: Under the same outdoor conditions, the second trial reached 51°C, thirteen degrees higher than the first.
The revision states the controlled condition and quantifies the difference clearly.

96. TACHS-WE3-046 - Answer A: By the time the bus arrived, we had waited for twenty minutes.
The past perfect correctly describes waiting that occurred before another past event.
97. TACHS-WE3-047 - Answer A: The coach spoke privately with him and me.
The preposition "with" requires the objective pronouns "him" and "me."
98. TACHS-WE3-048 - Answer D: Line 4
The past-tense context requires "organized," not "organize."
99. TACHS-WE3-049 - Answer C: Line 3
Comparing two routes requires "more steeply," not "most steeply."
100. TACHS-WE3-050 - Answer D: Line 4
The plural subject "trucks" requires the plural verb "move."

Mathematics

101. TACHS-M3-001 - Answer B: 27

$45 \div 5 = 9$, and $9 \times 3 = 27$.

102. TACHS-M3-002 - Answer B: \$48

The discount is $0.25 \times \$64 = \16 , so the price is $\$64 - \$16 = \$48$.

103. TACHS-M3-003 - Answer C: $\frac{5}{8}$

$\frac{5}{8} = 0.625$, which is greater than 0.62, 0.61, and 0.60.

104. TACHS-M3-004 - Answer A: 7

Add 7 to obtain $4x = 28$, then divide by 4.

105. TACHS-M3-005 - Answer B: $5n + 1$

The sequence increases by 5, and $5(1) + 1$ gives the first term, 6.

106. TACHS-M3-006 - Answer C: -10

Substitute -4: $y = 3(-4) + 2 = -12 + 2 = -10$.

107. TACHS-M3-007 - Answer A: 20

The sum is 100 pages, and $100 \div 5 = 20$.

108. TACHS-M3-008 - Answer A: $\frac{1}{16}$

The probability on each draw is $\frac{3}{12} = \frac{1}{4}$, so $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.

109. TACHS-M3-009 - Answer C: 65°

Triangle angles total 180° , and $180^\circ - 48^\circ - 67^\circ = 65^\circ$.

110. TACHS-M3-010 - Answer D: 38 cm

The width is $84 \div 12 = 7$, so the perimeter is $2(12 + 7) = 38$.

111. TACHS-M3-011 - Answer D: 0.75 L

Since 1,000 milliliters equal 1 liter, $750 \div 1,000 = 0.75$ liter.

112. TACHS-M3-012 - Answer D: 2 hours 35 minutes

From 9:35 to 11:35 is two hours, and to 12:10 is 35 more minutes.

113. TACHS-M3-013 - Answer C: 3

Convert $2\frac{1}{3}$ to $\frac{7}{3}$, then multiply by $\frac{9}{7}$: $\frac{7}{3} \times \frac{9}{7} = 3$.

114. TACHS-M3-014 - Answer B: 42

The greatest common factor of 84 and 126 is 42, so 42 identical kits can be made.

115. TACHS-M3-015 - Answer A: $-3 \times 6 + 4$

Apply order of operations: $-3 \times 6 + 4 = -18 + 4 = -14$.

116. TACHS-M3-016 - Answer D: $x \leq 5$

Subtract 5 and divide by 3: $3x \leq 15$, so $x \leq 5$.

117. TACHS-M3-017 - Answer B: $y = 3x + 1$

Each y-value is one more than three times x, so the rule is $y = 3x + 1$.

118. TACHS-M3-018 - Answer C: 20

The middle values are 19 and 21, and their average is 20.

119. TACHS-M3-019 - Answer A: $\frac{3}{8}$

Three of the eight equal sectors are labeled A, so the probability is $\frac{3}{8}$.

120. TACHS-M3-020 - Answer D: 25%

The car group has $120 - 42 - 30 - 18 = 30$ students, and $30/120 = 25\%$.

121. TACHS-M3-021 - Answer C: 8 units

The points share the same y-coordinate, and $5 - (-3) = 8$ units.

122. TACHS-M3-022 - Answer D: 15 centimeters

The scale factor is $\frac{10}{6}$, so the corresponding side is $9 \times \frac{10}{6} = 15$ centimeters.

123. TACHS-M3-023 - Answer B: 55 square centimeters

Area is $\frac{1}{2} \times (8 + 14) \times 5 = 55$ square centimeters.

124. TACHS-M3-024 - Answer A: 14 feet

Four yards equal 12 feet, and $12 + 2 = 14$ feet.

125. TACHS-M3-025 - Answer D: 3.05 kilograms
Convert 650 grams to 0.65 kilogram, then add $2.4 + 0.65 = 3.05$ kilograms.
126. TACHS-M3-026 - Answer C: \$48
A 25% discount is \$16, so the sale price is $\$64 - \$16 = \$48$.
127. TACHS-M3-027 - Answer A: $19/20$
Divide first: $7/10 \div 2 = 7/20$; then $12/20 + 7/20 = 19/20$.
128. TACHS-M3-028 - Answer D: 6
Oats are 2 of 7 ratio parts, so $2/7 \times 21 = 6$ cups.
129. TACHS-M3-029 - Answer B: Multiply by 2, then add 1.
Applying this rule produces 9, 19, and 39 successively.
130. TACHS-M3-030 - Answer C: $x = 6$
Expanding gives $8x - 12 = 3x + 18$, so $5x = 30$ and $x = 6$.
131. TACHS-M3-031 - Answer D: $7 + 2m$
The fixed monthly charge is 7, and m rentals cost $2m$.
132. TACHS-M3-032 - Answer A: 24
The values total 120, and $120 \div 5 = 24$.
133. TACHS-M3-033 - Answer C: $1/25$
With replacement, the probability is $2/10 \times 2/10 = 4/100 = 1/25$.
134. TACHS-M3-034 - Answer B: 65°
Triangle angles total 180° , so $180 - 47 - 68 = 65^\circ$.
135. TACHS-M3-035 - Answer A: 40 cm
The width is $96 \div 12 = 8$, so the perimeter is $2(12 + 8) = 40$.
136. TACHS-M3-036 - Answer D: 3,500
Since 1 liter equals 1,000 milliliters, $3.5 \times 1,000 = 3,500$.
137. TACHS-M3-037 - Answer B: 2 hours 31 minutes
From 10:47 to 12:47 is two hours, plus 31 minutes to 1:18.
138. TACHS-M3-038 - Answer D: 60 liters
The added 18 liters equal $9/10 - 3/5 = 3/10$ of capacity, so capacity is $18 \div 3/10 = 60$.
139. TACHS-M3-039 - Answer B: 14
First, $4.8 \div 0.6 = 8$ and $3 \times 2 = 6$; then $8 + 6 = 14$.
140. TACHS-M3-040 - Answer A: 15%
The increase is 12, and $12 \div 80 = 0.15 = 15\%$.
141. TACHS-M3-041 - Answer B: 9 cm
With width w , $2w + 2(w + 5) = 46$, so $4w = 36$ and $w = 9$.
142. TACHS-M3-042 - Answer C: 35
Sixty minutes contains five 12-minute intervals, so the machine produces $5 \times 7 = 35$ parts.
143. TACHS-M3-043 - Answer D: 46
Five days must total $5 \times 42 = 210$; the first four total 164, leaving 46.
144. TACHS-M3-044 - Answer C: $1/22$
Without replacement, the probability is $3/12 \times 2/11 = 6/132 = 1/22$.
145. TACHS-M3-045 - Answer A: Both sets have the same mean, but Q has a greater range.
Both totals are 50, while P's range is 8 and Q's range is 16.
146. TACHS-M3-046 - Answer A: 8
Exterior angles total 360° , so the polygon has $360 \div 45 = 8$ sides.
147. TACHS-M3-047 - Answer C: 113.04 cm^2
Area is $\pi r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$.
148. TACHS-M3-048 - Answer D: 158 cm^2
Surface area is $2(8 \times 5 + 8 \times 3 + 5 \times 3) = 2(79) = 158$.
149. TACHS-M3-049 - Answer C: 24 miles
The rate is $18 \div 1.5 = 12$ miles per hour, and $12 \times 2 = 24$.

150. TACHS-M3-050 - Answer B: 1.5 L

Six batches need $6 \times 250 = 1,500$ milliliters, which equals 1.5 liters.

Ability

151. TACHS-A3-001 - Answer A: ●●

Moving right adds one identical shape, while moving down changes each open circle to a filled circle.

152. TACHS-A3-002 - Answer C: ↑

Rotating the bottom-left arrow 90° clockwise changes ← to ↑.

153. TACHS-A3-003 - Answer B: ☆+

Each third cell joins the first and second symbols in their original left-to-right order.

154. TACHS-A3-004 - Answer D: filled ◆ above open △

The two shapes keep their forms but exchange top and bottom positions across the row.

155. TACHS-A3-005 - Answer B: ○

Alternation requires an open circle after the filled circle in both the final row and column.

156. TACHS-A3-006 - Answer C: Near both upper corners

Unfolding reflects the upper-right punch across the vertical centerline to the upper-left.

157. TACHS-A3-007 - Answer B: Four holes, one in each quadrant, symmetric across both centerlines

Reversing two perpendicular half-folds reflects the punch across both centerlines, producing four symmetric holes.

158. TACHS-A3-008 - Answer A: A diamond-shaped opening centered on the horizontal centerline

A triangular cut on the fold reflects into a matching triangle, forming a diamond-shaped opening.

159. TACHS-A3-009 - Answer A: Two holes reflected across that diagonal

Reversing one diagonal fold creates one mirror image across the fold line.

160. TACHS-A3-010 - Answer C: ◇◆◇

Each figure has matching open outer shapes surrounding the filled version of the same shape.

161. TACHS-A3-011 - Answer D: ◇ inside △

Like every example, this arrangement places one open shape inside a different open shape.

162. TACHS-A3-012 - Answer D: ← ↑ →

Each arrow turns 90° clockwise from the previous arrow: ← becomes ↑, then →.

163. TACHS-A3-013 - Answer D: ▲ ▲ ▲

Five triangles minus two triangles leaves three, and the rule changes the remaining shapes to filled triangles.

164. TACHS-A3-014 - Answer B: Left △; Right ◇

Turning the vertical pair counterclockwise places the top triangle on the left and the bottom diamond on the right.

165. TACHS-A3-015 - Answer A: △△

The third row must continue the shifted count pattern 3, 1, 2, and the last cell is open.

166. TACHS-A3-016 - Answer C: ●■

Applying both independent changes fills the circle and the square, producing ●■.

167. TACHS-A3-017 - Answer B: Three holes in a horizontal row, one in each panel

The three stacked vertical panels each receive the punch, producing one top-position hole in every panel.

168. TACHS-A3-018 - Answer D: Four holes, one near each corner

Each punch reflects across the vertical fold, producing matching upper and lower pairs near all four corners.

169. TACHS-A3-019 - Answer C: Six holes in a 2 × 3 arrangement

The horizontal fold doubles the three vertical layers, producing six reflected holes in two rows of three.

170. TACHS-A3-020 - Answer A: Two matching semicircular notches, one at the midpoint of each outer vertical edge

The stacked original outer edges are cut together, so unfolding places matching notches on the left and right edges.

171. TACHS-A3-021 - Answer A: □ △ ▲ ■

Each arrangement shows two open shapes followed by their filled counterparts in reverse order.

172. TACHS-A3-022 - Answer C: □ ▲ ▲

Each arrangement begins with one open shape and ends with two filled copies of a different shape.

173. TACHS-A3-023 - Answer D: ← ◇ ◆

Each arrangement has one arrow followed by an open shape and then that shape's filled counterpart.

174. TACHS-A3-024 - Answer B: | / -

Each valid group contains exactly one vertical, one horizontal, and one forward-diagonal line.

175. TACHS-A3-025 - Answer B: ■
Across each row, the open shape changes to its filled counterpart without changing form.
176. TACHS-A3-026 - Answer A: ↑
Moving right or down rotates each arrow 90° clockwise, so left becomes up.
177. TACHS-A3-027 - Answer A: △△△
One matching open shape is added from the first cell to the second in each row.
178. TACHS-A3-028 - Answer B: ◇△
The two shapes exchange left and right positions across each row.
179. TACHS-A3-029 - Answer C: △☆
The third cell combines the first and second cells in their original order.
180. TACHS-A3-030 - Answer D: Two holes near the upper-right and lower-right corners
Reversing the horizontal fold reflects the hole vertically while preserving its right-side position.
181. TACHS-A3-031 - Answer B: Four holes, one in each quadrant, reflected across both centerlines
Two perpendicular folds create four reflections symmetric across both centerlines.
182. TACHS-A3-032 - Answer A: One diamond-shaped opening above the center
A triangular cut on the fold reflects into a matching triangle, forming one diamond-shaped opening.
183. TACHS-A3-033 - Answer C: Two holes mirrored across the upper-left-to-lower-right diagonal
Reversing the single diagonal fold creates two locations reflected across that diagonal.
184. TACHS-A3-034 - Answer B: ◇ ◆ ◇
Each arrangement has matching open outer shapes surrounding their filled counterpart.
185. TACHS-A3-035 - Answer D: ← ↑ →
Each successive arrow rotates 90° clockwise, so left, up, right follows the shared rule.
186. TACHS-A3-036 - Answer C: □ inside △; △ inside ◇
Each figure forms a nested chain of three different open shapes.
187. TACHS-A3-037 - Answer D: Top-left ●; top-right ■; bottom-left ■; bottom-right ●
Matching symbols occupy opposite corners, producing diagonal symmetry.
188. TACHS-A3-038 - Answer B: ■ above ●
Moving right swaps top and bottom positions; moving down changes each open shape to its filled counterpart.
189. TACHS-A3-039 - Answer C: □□+
Moving right adds a plus sign, while moving down changes one circle into two squares.
190. TACHS-A3-040 - Answer A: ◇+
In each row, the third cell combines the first and second symbols in the same order.
191. TACHS-A3-041 - Answer D: Left ■; Right □
Across each row, the two symbols exchange positions while keeping their forms and fill states.
192. TACHS-A3-042 - Answer B: Four holes, one near each corner
Reversing perpendicular folds reflects the punch across both centerlines, producing one hole near each corner.
193. TACHS-A3-043 - Answer A: One diamond-shaped opening below the center
The cut lies on the fold, so its reflected triangular half forms one diamond-shaped opening.
194. TACHS-A3-044 - Answer C: Two holes mirrored across the vertical centerline
One vertical fold creates two horizontally reflected holes; the centerline position prevents additional vertical separation.
195. TACHS-A3-045 - Answer D: Two matching circular openings reflected across the vertical centerline
The cut on the horizontal fold becomes one circle, and reversing the vertical fold creates its mirror.
196. TACHS-A3-046 - Answer A: ◇ | ◆
Each valid pair shows an open shape followed by the filled form of the same shape.
197. TACHS-A3-047 - Answer C: ← ↑ →
Each arrow rotates 90° clockwise from the one before it.
198. TACHS-A3-048 - Answer B: □ inside ☆; ■ outside ☆
Each group places an open shape inside a boundary and its filled counterpart outside that same boundary.
199. TACHS-A3-049 - Answer D: Top: ■; Center: □; Bottom: ■
Each valid arrangement has identical top and bottom symbols with a different center symbol.

200. TACHS-A3-050 - Answer D: ☆ + × ☆

Each valid group repeats its first symbol in fourth position while using two different middle symbols.